IKK Theory → VSIM Kernel Bridge

Mapping Inverse Kaluza-Klein Scale Symmetry
onto the VSEPR-SIM Identity Layer

This document is the formal bridge between the Inverse Kaluza-Klein (IKK) theoretical framework (Documents I–III) and the `IDENTITY/` header library in VSEPR-SIM v5.2.0. Each theoretical construct is mapped to its direct C++ counterpart, with design rationale and the specific open variables that remain formally undetermined.

VSEPR-SIM Theoretical Division

v5.2.0 — WO-SM-IDENTITY

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§1 Overview and Purpose

The Inverse Kaluza-Klein (IKK) framework (Doc I) proposes that the standard quantum wavefunction $\Psi(\mathbf{r}, t)$ is a *scale-projected shadow* of a fuller identity structure $\mathcal{I}$ defined over a scale ladder $\{S_0, S_1, \ldots, S_D\}$. The observed 3D spatial behaviour is the S_3-level projection; corrections at higher scales produce residuals that connect to dark-sector and quantum-gravity phenomenology.

VSEPR-SIM does not implement a wavefunction. It is a deterministic atomistic simulation engine. The IKK framework provides three tools directly applicable to the VSIM kernel at the subatomic annotation layer:

1. **Scale ladder** — a principled way to label which interactions are active for each particle type (IKK Doc I §2, Doc II §B).
2. **Hidden-active condition** — detection of particles whose net-zero observable channel conceals nonzero internal structure (IKK Doc I §3.3, Doc II §D).
3. **Identity anchor and fuzz** — the weighted center of an identity bundle and its spatial spread (IKK Doc I §2.2–2.3).

The SM-channel identity layer (`include/IDENTITY/`) is the correct integration point for all three. The Bridge65 gate remains in force: subatomic energy-regime activation requires its own work order. The IKK additions are *annotation enrichments* only — they do not modify classical MD positions or forces.

§2 Scale Ladder Mapping

2.1 IKK Scale Ladder (Doc I §2.1, Doc II §B)

S_n	IKK DOF	Mediator	Obs. limit	VSIM interpretation
S_0	Binary existence $E \in \{0,1\}$	—	Planck	Particle exists in plasma
S_1	Linear vector	—	Sub-nuclear	Reserved (not yet used)
S_2	Relational / angular	Gluon	Nuclear (∼1 fm)	Color charge, Cornell force
S_3	Spatial / binding	Photon	Atomic (∼1 Å)	EM channel, Coulomb
S_4	Time-ordered	W, Z bosons	$\sim 10^{-18}$ m	Weak channel (reserved)
S_5	Mass-energy curvature	Graviton	Cosmological	Not modeled
S_D	Dark-scale	X_D (unknown)	$> L_{\mathrm{EM}}$	Not modeled

Table 1: IKK scale ladder mapped to VSIM particle annotations.

2.2 C++ Implementation: `ScaleLevel` and `scale_mask`

```
enum class ScaleLevel : uint8_t {
  S0 = 0,  // existence
  S1 = 1,  // vector (reserved)
  S2 = 2,  // color/gluon
  S3 = 3,  // EM/spatial
  S4 = 4,  // weak/time-ordered
  S5 = 5,  // curvature (not modeled)
  SD = 6,  // dark (not modeled)
};

inline constexpr uint8_t scale_bit(ScaleLevel s) noexcept { ... }
inline bool scale_active(uint8_t mask, ScaleLevel s) noexcept { ...
   }
```

Listing 1: ScaleLevel enum and helpers in identity_matrix.hpp

Each `ParticleIdentity` carries a `scale_mask` bitmask. The convenience constructors set defaults reflecting physics:

Particle archetype	`scale_mask` bits
Electron / positron	$S_0 + S_3 + S_4$ (EM + weak)
Quark	$S_0 + S_2 + S_3$ (color + EM)

§3 Identity Bundle and Anchor

3.1 IKK Theory (Doc I §2.2)

A particle is an identity bundle:

$$\mathcal{I}_i(t) = \{c_{i1}, c_{i2}, \ldots, c_{im}\} \tag{1}$$

with identity anchor:

$$\mathcal{A}_i(t) = \frac{\sum_k \alpha_{ik}\, c_{ik}(t)}{\sum_k \alpha_{ik}} \tag{2}$$

and fuzz radius:

$$r_f^2 = \frac{\sum_k \alpha_k \left\| P_{3D}(c_k) - \vec{r}^{\,\text{obs}} \right\|^2}{\sum_k \alpha_k} \tag{3}$$

The fuzz ratio:

$$\Phi_S = \frac{r_f}{\ell_S} \qquad \Phi_S \ll 1 \implies \text{point-like at scale } S \tag{4}$$

3.2 VSIM Mapping

VSIM particles are point-like at S_3 (classical MD positions are exact). The `r_fuzz` field on `ParticleIdentity` stores a caller-updated dimensionless spread proxy that can be populated when the particle is participating in a QGP substep or a pair interaction with significant kinematic spread. The `fuzz_ratio()` helper in `identity_matrix.hpp` converts this to the IKK Φ_S value.

```
// Phi_S = r_fuzz / ell_S
inline double fuzz_ratio(double r_fuzz, ScaleLevel s,
                         double ell_S3_ang = 1.0,
                         double ell_S2_fm  = 1.0e-5);
```

Listing 2: fuzz_ratio() in identity_matrix.hpp

Remark 3.1. The IKK identity anchor (2) is a weighted mean of multi-channel component values, not a simple 3D position. In VSIM the 3D position is held by the classical MD layer (`QCDParticle::pos`); the identity anchor is the abstract center of the *annotation* layer and is not directly observable. The two are related by $\vec{r}^{\,\text{obs}} = P_{3D}(\mathcal{A}_i)$.

§4 Hidden-Active Condition

4.1 Theory (Doc I §3.3, Doc II §D)

For a component vector $\mathbf{c}_j \in \mathbb{R}^m$ with projection weight $\boldsymbol{\alpha}$ and weight matrix $\mathbf{W} \succeq 0$:

$$I_j = \boldsymbol{\alpha}^\top \mathbf{c}_j \tag{5}$$

$$H_j = \mathbf{c}_j^\top \mathbf{W}\, \mathbf{c}_j \tag{6}$$

The **hidden-active condition**:

$$\boxed{I_j = 0, \quad H_j > 0} \tag{7}$$

Canonical example: neutron-like charge balance.
A neutral baryon composed of (up, down, down) quarks has:

$$I_{\text{charge}} = +\tfrac{2}{3} - \tfrac{1}{3} - \tfrac{1}{3} = 0 \tag{8}$$

$$H_{\text{charge}} = \left(\tfrac{2}{3}\right)^2 + \left(\tfrac{1}{3}\right)^2 + \left(\tfrac{1}{3}\right)^2 = \tfrac{2}{3} \neq 0 \tag{9}$$

The particle is charge-neutral at the K_{EM} projection level but carries nonzero internal charge intensity visible to higher-scale projections. This is not an approximation; it is a structural consequence of packing.

4.2 VSIM Implementation

Three hidden-intensity channels are tracked per `ParticleIdentity`:

```cpp
double H_charge {0.0};    // hidden charge intensity  H = sum q_k^2
double H_color  {0.0};    // hidden color intensity
double H_spin   {0.0};    // hidden spin intensity
```

Listing 3: Hidden intensity fields in particle_identity.hpp

The method `compute_hidden_intensity()` populates all three using the single-particle or quark-triplet proxy:

```cpp
// H_charge for elementary particle: H = q^2
// H_charge for baryon proxy (isospin bookkeeping):
//    n_up * (2/3)^2 + n_down * (1/3)^2
void compute_hidden_intensity() noexcept;

// Query: net-zero charge but H_charge > 0?
bool hidden_active_charge(double tol = 1e-9) const noexcept;
```

Listing 4: compute_hidden_intensity() key logic

`evaluate_pair()` in `interaction_candidate.hpp` populates the hidden-active flags for each member of the pair:

Flag	Meaning
`a_hidden_active_charge`	particle a: $I_{\text{charge}} = 0$, $H_{\text{charge}} > 0$
`b_hidden_active_charge`	particle b
`a_hidden_active_color`	particle a: color-neutral but $H_{\text{color}} > 0$
`b_hidden_active_color`	particle b
`pair_has_hidden_charge_activity`	logical OR of the charge flags

These flags are available to the live event logger and to any future kernel that needs to distinguish a "truly neutral" particle from a "net-zero-but-internally-structured" one.

§5 Matrix Packing Chain

5.1 IKK Theory (Doc II §C)

The quark component matrix:

$$\mathbf{C}_q = \begin{bmatrix} q_1 & c_1 & \eta_1 \\ q_2 & c_2 & \eta_2 \\ q_3 & c_3 & \eta_3 \end{bmatrix} \in \mathbb{R}^{3\times 3} \qquad \text{(charge, color, helicity columns)} \tag{10}$$

packed to particle identity row-vector:

$$\mathbf{I}_p = \Pi_q(\mathbf{C}_q) \in \mathbb{R}^{1\times 3} \qquad [Q_p,\ M_p,\ S_p] \tag{11}$$

The recursive packing ladder (Doc II eq. C.4):

$$\underbrace{\text{quarks}}_{S_2} \xrightarrow{\Pi_2} \underbrace{\text{particles}}_{S_3} \xrightarrow{\Pi_3} \underbrace{\text{atoms}}_{S_3} \xrightarrow{\Pi_3} \underbrace{\text{molecules}}_{S_3} \xrightarrow{\Pi_4} \underbrace{\text{materials}}_{S_4} \tag{12}$$

5.2 VSIM Mapping

The 3×3 component matrix $\mathbf{C}_q$ corresponds to the `Z3` field of `ParticleIdentity`. For quarks, the three rows hold one quark each; for mesons or baryons, the whole structure represents the internal composition. At present `Z3` is initialised to the identity matrix; the packing operator Π_q is a future kernel.

The entropy and complexity measures (Doc II §C.3–C.4):

$$\mathcal{S}_n = k_B \ln \Omega(\mathbf{I}^{(n)}) \tag{13}$$

$$\mathcal{C}_n = f(\mathbf{I}^{(n+1)}, \mathrm{Res}_n) \tag{14}$$

are not yet computed; they appear as open variables in the kernel development roadmap.

§6 Hidden Intensity: Z_2 and Z_3 as Inner-Structure Proxies

The two Z-matrices in `ParticleIdentity` encode the particle's field configuration at two resolutions:

Matrix	IKK analog	Current role
`Z2` (2×2)	Runtime identity state	kappa_Z (2x2 opposition)
`Z3` (3×3)	Component packing matrix $\mathbf{C}_q$	Future packing kernel

The `opposition()` function on `Z2` matrices produces `kappa_Z`, the leading term in the weighted kappa_total:

$$\kappa_Z = 1 - \frac{\langle Z_2^{(a)}, Z_2^{(b)} \rangle_F}{\|Z_2^{(a)}\|_F \|Z_2^{(b)}\|_F + \varepsilon} \tag{15}$$

This is the most direct numerical realisation of the IKK hidden-structure opposition concept at the 2D identity level.

The weight $w_Z = 0.40$ (40% of kappa_total) reflects the design choice that internal-structure opposition is the dominant interaction discriminator.

§7 Dark-Scale Residual and Open Variables

The IKK dark-sector equations (Doc I §6, Doc II §E) define:

$$\rho_{\text{dark}}(x, y, z;\, t) = \int \rho(x, y, z, w;\, t)\big[K_G(w) - K_{\text{EM}}(w)\big]\, dw \tag{16}$$

VSIM status: The dark coordinate w and both projection kernels K_G, K_{EM} are open variables (Doc II open-variable register §J). The `scale_mask` field with bit $S_D = 6$ is the forward compatibility hook; no dark-scale force or density computation is active.

A reference C++ implementation of `dark_density_residual()` using trapezoidal integration over a w-grid appears in Doc II §I (C++ code section) and in the `ikk::` namespace sketch distributed alongside that document.

§8 Projection Failure and State Loss

IKK Doc I §4 establishes that when projection P_S is non-injective, exact reconstruction is impossible. The VSIM `I_recoverable` field tracks the recoverable fraction of the identity state after each interaction:

$$\Delta I = I_{\text{out}} - I_{\text{in}} \quad \text{(signed change)} \tag{17}$$

$$L_{ij} = -\Delta I \quad \text{(state loss, } \geq 0) \tag{18}$$

State loss $L_{ij} > 0$ corresponds to IKK *destructive inverse packing*: information available at the higher scale ($\mathcal{I}$) is not recovered at the lower projection ($\mathcal{O}_{S_3}$). The annihilation gate requires $L_{ij} > L_{\text{ann}}$ (default 0.50) to confirm a full annihilation event.

Remark 8.1. The IKK framework connects this to the quantum no-cloning theorem and to the irreversibility of measurement. In VSIM this is operationalised as an interaction-record bookkeeping quantity, not a wavefunction collapse.

§9 Relation-Particle and Entanglement

IKK Doc I §5 / Doc II §G defines the relation-particle object:

$$\mathcal{R}_{AB} = [0,\ 0,\ 0,\ 0 \mid \mathbf{T}_{AB}] \qquad \mathbf{T}_{AB} \in \{-1,\, 0,\, +1\}^m \tag{19}$$

$\mathcal{R}_{AB}$ is a real, deterministic, non-spatiotemporal structure. Measurement at A and at B both read the same relation; no signal travels.

VSIM status: The trinary vector $\mathbf{T}_{AB}$ and its dynamics are open variables. The `InteractionCandidate` struct is the natural carrier for a relation-particle record when this is implemented; the `pair_has_hidden_charge_activity` flag is the first step toward identifying pairs that would carry a non-trivial $\mathbf{T}_{AB}$.

§10 Effective Proper Time

The IKK extended proper-time element (Doc I §7, Doc II §F):

$$d\tau_{\text{eff}} = dt\sqrt{1 - \frac{v^2}{c^2} + \frac{2\Phi_G}{c^2} + \chi_w(w)} \tag{20}$$

reduces to standard SR proper time when $\chi_w = 0$.

VSIM status: VSIM is a classical MD engine. No relativistic proper time is computed. The $T_A|B$ dual-timescale architecture (WO-63A) provides an operational time-dilation analog for QCD substeps (τ_{scale}), but this is a scale-bridging approximation, not the IKK dark-scale correction. The `scale_mask` bit S_4 (weak / time-ordered) is the forward hook for proper-time corrections if the engine is ever extended to relativistic integration.

§11 Code Reference: Files and Namespaces

File	IKK concept implemented
`include/IDENTITY/identity_matrix.hpp`	`ScaleLevel`, `scale_bit/active`, `fuzz_ratio()`, `hidden_intensity_diagonal()`, `quark_charge_hidden_intensity()`
`include/IDENTITY/particle_identity.hpp`	`ParticleIdentity`: `scale_mask`, `r_fuzz`, `H_charge/color/spin`, `compute_hidden_intensity()`, `hidden_active_charge()`
`include/IDENTITY/interaction_channel.hpp`	Conservation gates $G_E \cdots G_s$, kappa scoring, nine-channel row
`include/IDENTITY/interaction_candidate.hpp`	`evaluate_pair()`: hidden-active flags `a/b_hidden_active_charge/color`, `pair_has_hidden_charge_activity`
`include/IDENTITY/identity.hpp`	Umbrella header
`include/COLLISION/qcd_force.hpp`	`qcd_accumulate_forces_sm()`: SM-gated Cornell wrapper
`include/IKK/kernel_v4.hpp`	`ikk::KernelV4`: two-channel bilinear projection (Doc III); `ikk::KernelV4Multi<N_CHAN,N_STATE>`: generalised engine; `ikk::mixing_matrix_2ch()`: standard $\mathbf{M}_2$

All `IDENTITY/` code is in the `vsepr::identity` namespace. The bilinear projection kernel lives in the `ikk::` namespace (`include/IKK/kernel_v4.hpp`), separate from the identity layer, to keep the radial-density engine independent of particle bookkeeping.

§12 Open Variables Register (VSIM perspective)

§13 Summary

The IKK scale-symmetry framework provides the VSEPR-SIM IDENTITY layer with:

1. A principled scale-ladder labeling system (`ScaleLevel` enum, `scale_mask` bitmask) that connects VSIM particle types to their active force channels.
2. The hidden-active condition ($I_j = 0$, $H_j > 0$) implemented as `H_charge/color/spin` fields, `compute_hidden_intensity()`, and hidden-active flags on `InteractionCandidate`.
3. The fuzz proxy (`r_fuzz`) and `fuzz_ratio()` helper that operationalise $\Phi_S = r_f/\ell_S$ for future spatial-spread tracking.
4. The quark-component-matrix slot (`Z3`) as the target for the future packing operator Π_2.

Variable	IKK source	VSIM status
$\chi_w(w)$	Doc I §7 / Doc II F.5	Dark-scale metric correction. Not modeled. `scale_mask` S_D is the hook.
$K_G(w)$, $K_{\mathrm{EM}}(w)$	Doc I §6 / Doc II E.3	Projection kernels. C++ skeleton in Doc II §I. Not active.
Π_n	Doc II B.3	Packing operator. `Z3` is the target matrix. Axiom only.
$\mathbf{T}_{AB}$	Doc II G.2	Trinary relation vector. `InteractionCandidate` is the carrier. Not computed.
$\mathbf{W}$	Doc II D.3	Hidden-intensity weight matrix. Diagonal $\mathbf{W} = \mathbf{I}$ assumed currently.
$\hat{H}_w$	Doc I §3.2	Dark-scale Hamiltonian. No quantitative predictions without this.
$\mathcal{S}_n$, $\mathcal{C}_n$	Doc II C.3–C.4	Packing entropy and complexity. Roadmap item.

Table 2: Open variables from IKK theory and their VSIM status.

5. A documented open-variable register linking every undefined IKK quantity to the VSIM code location where it will eventually be implemented.

The bridge does not modify the classical MD layer. All additions are annotation-only and live entirely within the `vsepr::identity` namespace, respecting the Bridge65 separation doctrine.

Reading Path

1. **IKK Doc I** — theoretical foundation and Schrödinger alternative
2. **IKK Doc II** — scale axioms, matrix packing chain, hidden intensity, dark matter kernel equations, Python and C++ prototypes
3. **IKK Doc III** — Kernel V4 bilinear projection form
4. **section_sm_identity_layer** — full SM-channel IDENTITY layer documentation (conservation gates, κ_{total}, ΔI, live event types)
5. **This document** — IKK → VSIM kernel mapping